

KD-VOCODEC: A LOW-COMPLEXITY MODEL FOR JOINT SPEECH CODING AND ENHANCEMENT USING KNOWLEDGE DISTILLATION

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ABSTRACT

The advancement of speech interfaces operating in resource-constrained environments drives the need for neural speech codecs that achieve a critical balance among computational efficiency, minimized bitrate, and low latency. These codecs must also maintain high perceptual quality in adverse acoustic environments, demonstrating robustness to real-world noise and reverberation. In this paper, we propose KD-Vocodec, a novel framework that tackles this challenge via feature-level knowledge distillation for joint speech coding and enhancement. Our approach employs a feature-level knowledge distillation framework to train a compact student encoder that directly reconstructs the intermediate representations of a pre-trained, high-fidelity teacher codec, aligning the student’s output features with the teacher’s clean-speech embeddings. KD-Vocodec secured 2nd place in Track 2 of the 2025 Low-Resource Audio Codec (LRAC) Challenge, demonstrating superior performance across multiple objective and subjective metrics under strict low-resource constraints. Our system operates at a low latency of 30 ms—below the track’s 50 ms requirement—with scalable bitrates from 1 to 6 kbps.

Index Terms— neural speech codec, knowledge distillation, speech enhancement, low-resource audio coding

1. INTRODUCTION

Neural audio codecs have emerged as a key technology in modern speech communication and processing, enabling efficient compression and reconstruction of audio signals under low-bitrate constraints while maintaining perceptual fidelity [1, 2, 3]. Unlike traditional signal processing based codecs, neural codecs leverage deep neural networks to learn compact and information-rich representations directly from data, capturing complex auditory structures that conventional approaches often fail to model. This data-driven paradigm allows them not only to achieve high-quality reconstruction at significantly lower bitrates but also to produce discrete latent representations that can be directly utilized for a wide range of downstream tasks, such as speech enhancement, recognition, and synthesis [4, 5, 6].

Among the most prevalent architectures for neural audio coding is the VQ-GAN framework [7], which employs adversarial training to optimize a VQ-VAE [8] for the discrete encoding and reconstruction of speech signals. For instance, SoundStream [9] and EnCodec [10] are two pioneering examples of CNN-based end-to-end neural codecs that operate directly on waveform inputs, setting new benchmarks for high-fidelity audio streaming and synthesis. Beyond waveform-input models, another line of research focuses on transform-domain neural codecs [11, 12, 13], which leverage spectral representations to achieve alternative trade-offs between computational efficiency and reconstruction quality.

While recent codecs [14, 15, 16] focus on aggressively reducing bitrate, they often overlook the critical requirement for low computational resources, making them difficult to deploy on resource-limited hardware [17, 18]. Furthermore, in real-world scenarios, input audio is often distorted by background noise and reverberation, resulting in degraded intelligibility and perceptual quality of the reconstructed speech. The efficacy of recent codec approaches under complex acoustic environments remains underexplored, highlighting the need for integrated speech enhancement mechanisms within the coding pipeline to improve robustness [9, 19]. To address this gap, the 2025 Low-Resource Audio Codec (LRAC) Challenge [20]¹ is proposed, aiming to advance neural codecs that operate under strict constraints on computational cost, bitrate, and latency. Specifically, Track 2 of the challenge encourages the integration of speech enhancement directly into the coding process to improve robustness in noisy conditions.

To address this critical challenge, we propose KD-Vocodec, a novel system that jointly optimizes speech coding and enhancement within a unified architecture. Our method begins by training a teacher codec on clean speech for high-fidelity transparent coding. Subsequently, a student encoder is trained through knowledge distillation (KD). Specifically, the student learns to map a noisy input waveform to the clean representation generated by the teacher encoder from the corresponding clean waveform. The final KD-Vocodec system is assembled by integrating the student encoder with the quantizer and

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¹<https://crowdsourcing.cisco.com/lrac-challenge/2025/>

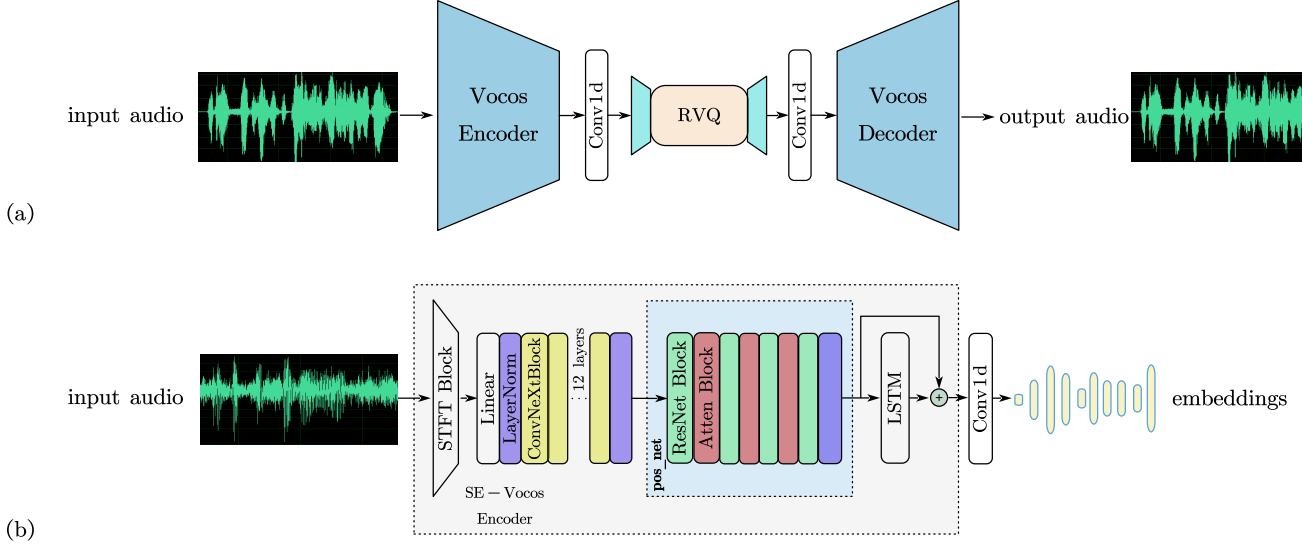


Fig. 1. Architecture of the proposed KD-Vocodec framework. (a) Overall pipeline of the teacher codec; (b) Architecture of the student encoder.

decoder from the teacher codec. Through this simple yet effective paradigm, we seamlessly convert a codec originally designed for transparent coding into one robust for noisy speech. The proposed KD-Vocodec achieved a final weighted score of 63.64 in the official LRAC Challenge crowdsourced listening tests, ranking 2nd in Track 2.

2. PROPOSED METHOD

Our proposed framework leverages feature-level KD to achieve joint speech coding and enhancement under strict latency and bitrate constraints. The system architecture, depicted in Fig.1, is built upon a VQ-GAN-based clean teacher codec. The overarching design employs a teacher-student paradigm wherein a student encoder is trained to replicate the intermediate representations of a pre-trained teacher encoder, facilitating the learning of clean features. We provide audio demos processed by KD-Vocodec under various acoustic conditions and bitrates on a public webpage².

2.1. Teacher codec architecture

We adopt Vocos [21] as the backbone of our teacher codec architecture, due to its superior performance in speech synthesis. Specifically, a mirrored variant of the Vocos structure is employed as the encoder-decoder backbone. The input waveform is first converted into a time-frequency representation via short-time Fourier transform (STFT). The complex spectrogram is split into magnitude and phase components, which are concatenated along the frequency dimension and fed into the network. This combined input is projected into a latent space with dimension D via a linear layer. The encoder consists of multiple ConvNeXt blocks [22], aiming to extract

deep hierarchical features. Each block contains a 1D depthwise convolution with weight normalization, followed by a pointwise convolution. To ensure strict causality and avoid algorithmic delay, all temporal padding is causal. To enhance sequence modeling, ResNet blocks are incorporated. Inspired by WavTokenizer [23], a causal self-attention mechanism is inserted after the second ResNet block. The resulting features are passed to the quantizer, which uses a Residual Vector Quantizer (RVQ) [24] with 6 layers and gradient-based codebook updates. Linear projection layers are introduced both preceding and following the RVQ to alleviate the computational complexity of quantization and avoid redundant channel information. These projections first compress the encoded representations into a lower-dimensional latent subspace and subsequently reconstruct them to the original dimension.

The encoder configuration is as follows: STFT window size is 720 samples with a hop size of 240; hidden dimension D is 256; the encoder stack contains 12 layers, each with an expansion channel size of 896. The decoder mirrors the encoder’s structure but with reduced capacity to meet receiver-side computational constraints: D is 252, the number of layers is 4, and the expansion channel size is 256. The projection dimension for RVQ is set to 8.

2.2. Student encoder

The student encoder is designed by augmenting the baseline encoder with several key components to enhance its ability to derive clean and robust embeddings from distorted speech inputs. Specifically, causal self-attention modules are inserted after each ResNet block, except for the final one, to capture long-range contextual dependencies. In addition, a two-layer LSTM is placed immediately before the final convolutional

²<https://yangxu212.github.io/KD-Vocodec/>

layer to improve temporal sequence modeling, with a skip connection linking its input and output to facilitate gradient flow. All introduced modules are implemented in a frame-wise causal manner, ensuring consistent latency to the original encoder. All other architectural configurations of the student encoder are kept identical to those of the teacher encoder.

2.3. Discriminator

Given that the input to our model is derived from the STFT time-frequency representation, it is advantageous to employ a Multi-Scale STFT Discriminator (MSSTFTD) [10] to assess the reconstruction quality directly in the spectral domain. A set of window lengths [128, 256, 512, 1024, 2048] is used, and the hop length is fixed to one-fourth of the window length. Accordingly, we introduce adversarial training solely using the MSSTFTD to refine the output of the teacher codec.

2.4. Loss function

The training of the teacher codec utilizes a composite loss function within a GAN framework. The total generator loss, $L_{generator}$, is a weighted sum of multiple components: the multi-scale mel-spectrogram reconstruction loss L_{rec} [25], the generator adversarial loss L_g , the feature matching loss L_{feat} applied to the discriminator’s features, the codebook loss L_{code} , and the commitment loss L_c . It is formulated as:

$$L_{rec} = \|\mathcal{M}(x) - \mathcal{M}(\hat{x})\|_1 \quad (1)$$

$$L_g = \|1 - D(\hat{x})\|_2^2 \quad (2)$$

$$L_{feat} = 2 \sum_l \|D^l(x) - D^l(\hat{x})\|_1 \quad (3)$$

$$L_{generator} = \lambda_{rec} L_{rec} + \lambda_g L_g + \lambda_{feat} L_{feat} + \lambda_{code} \underbrace{\|\text{sg}[\mathbf{z}_e] - \mathbf{e}_k\|_2^2}_{L_{code}} + \lambda_c \underbrace{\|\mathbf{z}_e - \text{sg}[\mathbf{e}_k]\|_2^2}_{L_c} \quad (4)$$

In the above equations, x and $\hat{x}$ denote the target and reconstructed speech, respectively, $\mathcal{M}(\cdot)$ is the mel-spectrogram transform, $D(\cdot)$ is the discriminator output, $D^l(\cdot)$ represents the feature map of the l -th discriminator layer, $\mathbf{z}_e$ is the quantizer output, and $\mathbf{e}_k$ is the codebook vector. The multi-scale mel-spectrogram loss L_{rec} is computed using window length samples [32, 64, 128, 256, 512, 1024, 2048], with the hop length fixed at 1/4 of each window length. Each scale uses different mel bins of [5, 10, 20, 40, 80, 160, 320]. Loss weights are set as: $\lambda_{rec} = 15$, $\lambda_g = 2$, $\lambda_{feat} = 1$, $\lambda_{code} = 1$, $\lambda_c = 0.25$. The discriminator is trained separately with the adversarial loss L_d :

$$L_d = \|1 - D(x)\|_2^2 + \|D(\hat{x})\|_2^2. \quad (5)$$

For knowledge distillation in the student encoder, the loss combines the MSE and cosine distance between the teacher and student embeddings, weighted by 1.0 and 0.1, respectively.

3. EXPERIMENTS

3.1. Training data preparation

We trained our model on a large-scale speech dataset curated from high-quality speech samples obtained from the EARS, VCTK, Common Voice, LibriTTS, Multilingual LibriSpeech datasets, and DNS Challenge 5 dataset. All speech signals are resampled to 24 kHz. To extend the noise suppression and dereverberation capabilities of the model, we further constructed a noise data set that includes noise from the VCTK, WHAM, FSD50K, and FMA datasets, encompassing various noise types. During training of student encoder, each speech sample is combined with background noise with an 80% probability, where signal-to-noise ratios (SNRs) are uniformly distributed between -5 dB and 30 dB. For reverberation, we use room impulse responses (RIRs) from the Motus dataset, and each sample is augmented with reverberation with a probability of 50% during training. All training data follow the cleaning and preprocessing procedures defined in the challenge data preparation repository³.

3.2. Implementation details

Notably, our approach avoids using any pre-trained models throughout the training and inference pipeline. The training procedure consists of two distinct stages. The first stage involves training the teacher codec using a GAN-based reconstruction objective. This model is trained for 1000 epochs with a batch size of 128, using the AdamW optimizer with a cosine annealing learning rate scheduler. In the subsequent distillation stage, the student encoder is trained to replicate the teacher’s embeddings. This stage runs for 500 epochs with a batch size of 384, optimized by RAdam with an exponential decay scheduler.

3.3. Computational complexity

The proposed KD-Vocodec system operates with a total computational complexity of 1.28G multiply-accumulate operations per second (MACs/s) and contains 12.65M parameters. Specifically, the student encoder accounts for 981.79M MACs/s and 9.70M parameters, while the decoder requires 299.84M MACs/s and 2.95M parameters. The system maintains strict causality without look-ahead. Consequently, the algorithmic latency is determined solely by the 30-ms STFT analysis window at a 24 kHz sampling rate. The system supports variable bitrates via its RVQ module, where each quantizer layer provides approximately 1 kbps (using a 1024-codebook at 100 fps), allowing operational modes of 1 kbps (1-layer) and 6 kbps (6-layers).

3.4. Results

3.4.1. Ablation Study

To validate the effectiveness of our proposed student encoder architecture, we conducted a systematic ablation study focusing on two critical design choices: (1) the contribution of in-

³https://github.com/cisco-open/lrac_data_generation

Table 2. Subjective metrics comparison on blind test set

	Clean speech		Real-world speech in noise		Real-world speech reverb		Intelligibility in clean	Intelligibility in noise	Aggregate Score
	MUSHRA [0, 100]		MOS [1, 5]		MOS [1, 5]		DRT score [-100, 100]	DRT score [-100, 100]	
	1kbps	6kbps	1kbps	6kbps	1kbps	6kbps	1kbps	6kbps	
	10%	15%	10%	20%	10%	5%	10%	10%	
Baseline	21.26	60.06	1.30	2.25	1.39	2.32	75.86	68.21	38.52
UL-UNAS cascade	58.63	75.96	2.09	2.89	2.15	3.05	76.21	66.35	55.28
Proposed	63.66	81.10	2.33	3.18	2.57	3.57	84.46	72.01	63.64

Table 1. Ablation study on student encoder architectures.

Bitrate	Test Set	Model	ScoreQ-ref↓	UTMOS↑	Sheet-SSQA↑	PESQ↑	Audiobox AE-CE↑
6 kbps	Clean	Base	0.160	3.717	3.934	3.143	5.778
		Pre-LSTM	0.156	3.737	3.944	3.178	5.789
		ConvNeXt-only	0.160	3.725	3.963	3.170	5.780
		Post-LSTM	0.153	3.749	3.982	3.219	5.794
	Noisy	Base	0.430	3.316	3.585	2.171	5.358
		Pre-LSTM	0.409	3.354	3.641	2.178	5.405
		ConvNeXt-only	0.420	3.333	3.600	2.202	5.371
		Post-LSTM	0.396	3.366	3.665	2.225	5.431
	Reverb	Base	0.516	3.013	3.627	1.723	5.368
		Pre-LSTM	0.468	3.090	3.677	1.762	5.429
		ConvNeXt-only	0.499	3.060	3.624	1.769	5.377
		Post-LSTM	0.480	3.096	3.658	1.802	5.426
1 kbps	Clean	Base	0.393	3.254	3.619	1.909	5.490
		Pre-LSTM	0.386	3.248	3.614	1.916	5.507
		ConvNeXt-only	0.390	3.255	3.608	1.914	5.494
		Post-LSTM	0.378	3.265	3.659	1.941	5.506
	Noisy	Base	0.557	2.966	3.367	1.582	5.248
		Pre-LSTM	0.542	3.007	3.404	1.590	5.295
		ConvNeXt-only	0.543	2.989	3.397	1.603	5.255
		Post-LSTM	0.530	3.006	3.434	1.609	5.302
	Reverb	Base	0.648	2.685	3.329	1.390	5.154
		Pre-LSTM	0.622	2.734	3.360	1.407	5.181
		ConvNeXt-only	0.638	2.735	3.348	1.420	5.130
		Post-LSTM	0.619	2.749	3.386	1.430	5.190

corporating recurrent and attention components compared to simply increasing model depth, and (2) the impact of their relative ordering when both recurrent and attention modules are integrated into the architecture. Specifically, for the first design choice, we compared against a baseline where we added two more standard ConvNeXt Blocks to increase depth without recurrent or attention mechanisms, denoted as ConvNeXt-only. For the second design, we evaluated the performance of inserting the LSTM before or after the pos-net, denoted as Pre-LSTM and Post-LSTM configurations, respectively.

All models were evaluated on the open test set⁴ provided by the challenge, using key objective metrics—including ScoreQ_ref [26], UTMOS [27], Sheet-SSQA [28], PESQ [29], and Audiobox_Aesthetics.CE [30]. Crucially, all compared configurations were carefully designed to maintain nearly identical computational complexity in terms of MACs/s, ensuring a fair and persuasive comparison. The results are presented in Table 1.

The ablation results demonstrate that integrating recurrent and attention mechanisms yields clear advantages over merely increasing network depth, as these modules enhance the encoder’s ability to model long-range dependencies and context-sensitive acoustic structures. Moreover, the consistent superiority of the Post-LSTM configuration across all test conditions indicates that temporal modeling is most effective when applied after contextual aggregation. This configura-

tion allows the LSTM to refine temporally coherent representations on contextually enriched features, effectively stabilizing frame-level variations before quantization and leading to more robust and discriminative latent representations.

3.4.2. Comparison with the challenge baseline model

The performance of our proposed KD-Vocodec system is evaluated against the official baseline for Track 2 of the LRAC Challenge on the blind test set⁵. For comparison, we also include a cascaded baseline that applies UL-UNAS [31], a lightweight speech enhancement model, as a preprocessing front-end. Although intuitive, this cascaded design introduces residual distortions that mismatch the codec’s clean-speech-optimized training domain, degrading reconstruction quality. In contrast, our feature-level knowledge distillation enables the student encoder to reconstruct the teacher’s clean features before quantization, effectively purifying the feature space and ensuring that codebook capacity is devoted to speech content rather than noise.

As shown in Table 2, crowdsourced listening tests demonstrate that KD-Vocodec consistently surpasses both baselines across all evaluation metrics, achieving higher MUSHRA [32] scores for clean quality, better DMOS for robustness, and improved DRT [33] for intelligibility. These results validate the effectiveness of our distillation-based framework for robust speech coding with inherent enhancement capability under real-world conditions.

4. CONCLUSION

This paper presents KD-Vocodec, a novel framework for joint speech coding and enhancement, detailed as our entry for Track 2 of the 2025 LRAC Challenge. Experimental results confirm that the system consistently outperforms the baseline under various acoustic scenarios. In the final challenge evaluation, KD-Vocodec earned 2nd place, which is a notable achievement that underscores the practical potential of our method for deploying high-quality, robust speech codecs in resource-limited environments.

5. ACKNOWLEDGEMENTS

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⁴<https://github.com/cisco/multilingual-speech-testing/tree/main/LRAC-2025-test-data/open-test-set> ⁵<https://github.com/cisco/multilingual-speech-testing/tree/main/LRAC-2025-test-data/blind-test-set>

6. REFERENCES

- [1] Yuhao Zhao, Maoshen Jia, Jiawei Ru, Lizhong Wang, and Liang Wen, "Tff-codec: A high fidelity end-to-end neural audio codec," *Circuits, Systems, and Signal Processing*, vol. 44, no. 6, pp. 4013–4032, 2025.
- [2] Xiao-Hang Jiang, Yang Ai, Rui-Chen Zheng, Hui-Peng Du, Ye-Xin Lu, and Zhen-Hua Ling, "Mdctcodec: A lightweight mdct-based neural audio codec towards high sampling rate and low bitrate scenarios," in *2024 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2024, pp. 540–547.
- [3] Hubert Siuzdak, Florian Grötschla, and Luca A Lanzendörfer, "Snac: Multi-scale neural audio codec," *arXiv preprint arXiv:2410.14411*, 2024.
- [4] Jixun Yao, Hexin Liu, Chen Chen, Yuchen Hu, EngSiong Chng, and Lei Xie, "Gense: Generative speech enhancement via language models using hierarchical modeling," *arXiv preprint arXiv:2502.02942*, 2025.
- [5] Tianrui Wang, Long Zhou, Ziqiang Zhang, Yu Wu, Shujie Liu, Yashesh Gaur, Zhuo Chen, Jinyu Li, and Furu Wei, "Viola: Conditional language models for speech recognition, synthesis, and translation," *IEEE/ACM transactions on audio, speech, and language processing*, 2024.
- [6] Chengyi Wang, Sanyuan Chen, Yu Wu, Ziqiang Zhang, Long Zhou, Shujie Liu, Zhuo Chen, Yanqing Liu, Huaming Wang, Jinyu Li, et al., "Neural codec language models are zero-shot text to speech synthesizers," *arXiv preprint arXiv:2301.02111*, 2023.
- [7] Patrick Esser, Robin Rombach, and Bjorn Ommer, "Taming transformers for high-resolution image synthesis," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 12873–12883.
- [8] Aaron Van Den Oord, Oriol Vinyals, et al., "Neural discrete representation learning," *Advances in neural information processing systems*, vol. 30, 2017.
- [9] Neil Zeghidour, Alejandro Luebs, Ahmed Omran, Jan Skoglund, and Marco Tagliasacchi, "Soundstream: An end-to-end neural audio codec," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 495–507, 2021.
- [10] Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi, "High fidelity neural audio compression," *arXiv preprint arXiv:2210.13438*, 2022.
- [11] Zhihao Du, Shiliang Zhang, Kai Hu, and Siqi Zheng, "Funcodec: A fundamental, reproducible and integrable open-source toolkit for neural speech codec," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 591–595.
- [12] Yang Ai, Xiao-Hang Jiang, Ye-Xin Lu, Hui-Peng Du, and Zhen-Hua Ling, "Apcodec: A neural audio codec with parallel amplitude and phase spectrum encoding and decoding," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 3256–3269, 2024.
- [13] Xue Jiang, Xiulian Peng, Yuan Zhang, and Yan Lu, "Disentangled feature learning for real-time neural speech coding," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [14] Detai Xin, Xu Tan, Shinnosuke Takamichi, and Hiroshi Saruwatari, "Bigcodec: Pushing the limits of low-bitrate neural speech codec," *arXiv preprint arXiv:2409.05377*, 2024.
- [15] Hanzhao Li, Liumeng Xue, Haohan Guo, Xinfu Zhu, Yuanjun Lv, Lei Xie, Yunlin Chen, Hao Yin, and Zhifei Li, "Single-codec: Single-codebook speech codec towards high-performance speech generation," *arXiv preprint arXiv:2406.07422*, 2024.
- [16] Yong Ren, Tao Wang, Jiangyan Yi, Le Xu, Jianhua Tao, Chu Yuan Zhang, and Junzuo Zhou, "Fewer-token neural speech codec with time-invariant codes," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 12737–12741.
- [17] Liang Xu, Jing Wang, Jianqian Zhang, and Xiang Xie, "Lightcodec: A high fidelity neural audio codec with low computation complexity," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 586–590.
- [18] Sunghwan Ahn, Beom Jun Woo, Min Hyun Han, Chanyeong Moon, and Nam Soo Kim, "Hilcodec: High-fidelity and lightweight neural audio codec," *IEEE Journal of Selected Topics in Signal Processing*, 2024.
- [19] Ju Lin, Kaustubh Kalgaonkar, Qing He, and Xin Lei, "Speech enhancement for low bit rate speech codec," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 7777–7781.
- [20] Kamil Wojcicki, Yusuf Ziya Isik, Laura Lechler, Mansur Yesilbursa, Ivana Balić, Wolfgang Mack, Rafał Łaganowski, Guoqing Zhang, Yossi Adi, Minje Kim, et al., "Low-resource audio codec (Irac): 2025 challenge description," *arXiv preprint arXiv:2510.23312*, 2025.
- [21] Hubert Siuzdak, "Vocos: Closing the gap between time-domain and fourier-based neural vocoders for high-quality audio synthesis," *arXiv preprint arXiv:2306.00814*, 2023.
- [22] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11976–11986.
- [23] Shengpeng Ji, Ziyue Jiang, Wen Wang, Yifu Chen, Minghui Fang, Jialong Zuo, Qian Yang, Xize Cheng, Zehan Wang, Ruiqi Li, et al., "Wav-tokenizer: an efficient acoustic discrete codec tokenizer for audio language modeling," *arXiv preprint arXiv:2408.16532*, 2024.
- [24] Bing-Hwang Juang and A Gray, "Multiple stage vector quantization for speech coding," in *ICASSP'82. IEEE International Conference on Acoustics, Speech, and Signal Processing*. IEEE, 1982, vol. 7, pp. 597–600.
- [25] Rithesh Kumar, Prem Seetharaman, Alejandro Luebs, Ishaan Kumar, and Kundan Kumar, "High-fidelity audio compression with improved rvqgan," *Advances in Neural Information Processing Systems*, vol. 36, pp. 27980–27993, 2023.
- [26] Alessandro Ragano, Jan Skoglund, and Andrew Hines, "Scoreq: Speech quality assessment with contrastive regression," *Advances in Neural Information Processing Systems*, vol. 37, pp. 105702–105729, 2024.
- [27] Takaaki Saeki, Detai Xin, Wataru Nakata, Tomoki Koriyama, Shinnosuke Takamichi, and Hiroshi Saruwatari, "Utmos: Utokyo-sarulab system for voicemos challenge 2022," *arXiv preprint arXiv:2204.02152*, 2022.
- [28] Wen-Chin Huang, Erica Cooper, and Tomoki Toda, "Mos-bench: Benchmarking generalization abilities of subjective speech quality assessment models," *arXiv preprint arXiv:2411.03715*, 2024.
- [29] Antony W Rix, John G Beerends, Michael P Hollier, and Andries P Hekstra, "Perceptual evaluation of speech quality (pesq)-a new method for speech quality assessment of telephone networks and codecs," in *2001 IEEE international conference on acoustics, speech, and signal processing. Proceedings (Cat. No. 01CH37221)*. IEEE, 2001, vol. 2, pp. 749–752.
- [30] Andros Tjandra, Yi-Chiao Wu, Baishan Guo, John Hoffman, Brian Ellis, Apoorv Vyas, Bowen Shi, Sanyuan Chen, Matt Le, Nick Zacharov, et al., "Meta audiobox aesthetics: Unified automatic quality assessment for speech, music, and sound," *arXiv preprint arXiv:2502.05139*, 2025.
- [31] Xiaobin Rong, Dahan Wang, Yuxiang Hu, Changbao Zhu, Kai Chen, and Jing Lu, "Ul-unas: Ultra-lightweight u-nets for real-time speech enhancement via network architecture search," *arXiv preprint arXiv:2503.00340*, 2025.
- [32] Laura Lechler, Chamran Moradi, and Ivana Balić, "Crowdsourcing mushra tests in the age of generative speech technologies: A comparative analysis of subjective and objective testing methods," *arXiv preprint arXiv:2506.00950*, 2025.
- [33] Laura Lechler and Kamil Wojcicki, "Crowdsourced multilingual speech intelligibility testing," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 1441–1445.